

CIS4560 Term Project Tutorial



Authors: Isabella Jimenez, Jasmine Carvalho, Karla Salazar, Sherman Liang

Instructor: [Jongwook Woo](#)

Date: 12/13/2025

Lab Tutorial

Isabella Jimenez, Jasmine Carvalho, Karla Salazar, Sherman Liang,

ijimen48@calstatela.edu, jcarval2@calstatela.edu, ksalaz63@calstatela.edu,
sliang30@calstatela.edu

12/13/2025

Chicago Crime Data Analysis using Hadoop and Hive

Objectives

In this hands-on lab, you will learn how to:

- Upload the Chicago Crime dataset to HDFS.
- Create and clean Hive external tables in Beeline.
- Create summary tables for analysis (e.g., crime types, locations, years)
- Export the cleaned results into a CSV file for visualizations.

- Build visualizations using Excel and Power BI.

Platform Spec

- Hadoop Cluster Version: Hadoop 2.7.1.3.4.2.0-258
- Hive version: Hive 1.2.1000.2.4.2.0-258
- Number of nodes: 5
- HDFS Capacity: 390.2 GB total
- OCPUs: 6
- Memory: ~16 GB per node
- Storage: 80-100 GB per node
- CPU speed: 2.45 GHz
- CPU Model name: AMD EPYC 7763 64-Core Processor

```
Architecture:      x86_64
CPU op-mode(s):    32-bit, 64-bit
Byte Order:        Little Endian
CPU(s):            6
On-line CPU(s) list: 0-5
```

```
-bash-4.2$ hdfs dfs -df -h
Filesystem                                Size      Used Available Use%
hdfs://bigdaimn0.sub03291929060.trainingvcn.oraclevcn.com:8020 390.2 G  109.2 G   280.1 G   28%
```

Step 1: Download and Dataset to the Hadoop Server

This step involves manually downloading the CSV file from the original website and transferring it from your local computer to a Hadoop server directory, where it can be accessed and processed later.

1. Open Git Bash and log into the Hadoop cluster:

```
ssh ksalaz63@129.153.113.98
```

2. Create a working directory:

```
mkdir -p ~/chicago_project
exit
```

3. In a new Git Bash window, upload your dataset:

```
scp
"C:\Users\kesal\Downloads\Crimes_-_2001_to_Present_20251026.csv"
ksalaz63@129.153.113.98:~/chicago_project/
```

```
kesal@KSLAPTOP MINGW64 ~  
scp "C:\Users\kesal\Downloads\Crimes_-_2001_to_Present_20251026.csv" ksalaz63@129.153.113.98:~/chicago_project/  
ksalaz63@129.153.113.98's password:  
.:Users\kesal\Downloads\Crimes_-_2001_to_Present_20251026.csv      100% 2234MB   1.3MB/s   28:35
```

4. On the same Git Bash window, verify that the upload was successful.

```
cd ~/chicago_project  
ls -lh
```

Step 2: Upload Dataset to HDFS

This step involves transferring the dataset into the Hadoop Distributed File System (HDFS), which enables Hive to read the file and later allows us to perform queries on it.

1. Create HDFS directories:

```
hdfs dfs -mkdir /user/ksalaz63/chicago_crime
```

2. In a Git Bash window, reconnect to the Hadoop cluster:

```
ssh ksalaz63@129.153.113.98
```

```
hdfs dfs -put  
~/chicago_project/Crimes_-_2001_to_Present_20251026.csv  
/user/ksalaz63/chicago_crime/  
  
hdfs dfs -ls /user/ksalaz63/chicago_crime
```

```
$ ssh ksalaz63@129.153.113.98  
ksalaz63@129.153.113.98's password:  
Last login: Sat Nov  1 02:56:41 2025 from 76.94.185.213  
-bash-4.2$ ls -lh ~/chicago_project  
total 2.2G  
-rw-r--r-- 1 ksalaz63 ksalaz63 2.2G Nov  1 02:03 Crimes_-_2001_to_Present_20251026.csv  
-bash-4.2$ hdfs dfs -mkdir /user/ksalaz63/chicago_crime  
-bash-4.2$ hdfs dfs -put ~/chicago_project/Crimes_-_2001_to_Present_20251026.csv /user/ksalaz63/chicago_crime/  
-bash-4.2$ hdfs dfs -ls /user/ksalaz63/chicago_crime  
Found 1 items  
-rw-r--r-- 3 ksalaz63 hdfs 2342389441 2025-11-01 08:23 /user/ksalaz63/chicago_crime/Crimes_-_2001_to_Present_20251026.csv  
-bash-4.2$
```

Step 3: Connect to Hive using Beeline and Create Hive Tables

This step involves opening the Hive environment (via beeline), where we will create tables, clean data, and run queries.

1. In a new Git Bash window, reconnect to the Hadoop cluster:

```
ssh ksalaz63@129.153.113.98
```

2. Open Beeline, using the command:

```
beeline
```

3. Make sure your database is there and use your own database:

```
Show databases;  
Use ksalaz63;
```

4. Create the first external table, `chicago_raw`, using this:

```
DROP TABLE IF EXISTS chicago_raw;  
  
CREATE EXTERNAL TABLE IF NOT EXISTS chicago_raw (  
    id BIGINT,  
    case_number STRING,  
    `date` STRING,  
    block STRING,  
    iucr STRING,  
    primary_type STRING,  
    description STRING,  
    location_description STRING,  
    arrest BOOLEAN,  
    domestic BOOLEAN,
```

```

beat STRING,
district STRING,
ward INT,
community_area STRING,
fbi_code STRING,
x_coordinate DOUBLE,
y_coordinate DOUBLE,
year INT,
updated_on STRING,
latitude DOUBLE,
longitude DOUBLE,
location STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
STORED AS TEXTFILE
LOCATION '/user/ksalaz63/chicago_crime'
TBLPROPERTIES ("skip.header.line.count"="1");

```

5. Create a second external table, `chicago_cleaned`, with this:

```

DROP TABLE IF EXISTS chicago_cleaned;
CREATE EXTERNAL TABLE IF NOT EXISTS chicago_cleaned (
  id STRING,
  case_number STRING,
  `date` STRING,
  block STRING,
  iucr STRING,

```

```

primary_type STRING,
description STRING,
location_description STRING,
arrest STRING,
domestic STRING,
beat STRING,
district STRING,
ward STRING,
community_area STRING,
fbi_code STRING,
x_coordinate STRING,
y_coordinate STRING,
year STRING,
updated_on STRING,
latitude DOUBLE,
longitude DOUBLE,
location STRING
)
ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'
WITH SERDEPROPERTIES (
    "separatorChar" = ",",
    "quoteChar" = "\"",
    "escapeChar" = "\\"
)
STORED AS TEXTFILE
LOCATION '/user/ksalaz63/chicago_crime'
TBLPROPERTIES ("skip.header.line.count"="1");

```

6. Create a third table, `chicago_final`, that we will be using for visualizations:

```
DROP TABLE IF EXISTS chicago_final;

CREATE TABLE chicago_final AS

SELECT

    id,

    case_number,

    from_unixtime(unix_timestamp(`date`, 'MM/dd/yyyy hh:mm:ss a')) AS
    incident_timestamp,

    block,

    primary_type,

    description,

    location_description,

    arrest,

    domestic,

    beat,

    district,

    ward,

    community_area,

    year,

    latitude,

    longitude

FROM chicago_cleaned

WHERE latitude IS NOT NULL AND longitude IS NOT NULL;
```

7. Verify that you have the three tables you created:

```
show tables;
```

tab_name
baseball_salaries
building
chicago_cleaned
chicago_final
chicago_raw

8. Use these queries to verify that the data is being read properly:

***Note:** Some results might be different from the screenshot due to the new data being added.*

```
SELECT COUNT(*) AS total_records FROM chicago_final;
```

chicago_cleaned_id	chicago_cleaned_case_number	chicago_cleaned_arrest	chicago_cleaned_date	chicago_cleaned_block	chicago_cleaned_tucr	chicago_cleaned_primary_type	chicago_cleaned_description	chicago_cleaned_location_description	chicago_cleaned_x_coordinate	chicago_cleaned_y_coordinate	chicago_cleaned_year	chicago_cleaned_updated_on	chicago_cleaned_latitude	chicago_cleaned_longitude	chicago_cleaned_location
14001782	1151141	1850369	10/18/2025 12:00:00 AM	081XX S PULASKI RD	0610	BURGLARY	70	RESTAURANT	41.745293332	-87.721774867	2025	10/25/2025 03:43:27 PM	41.745293332	-87.721774867	(41.745293332, -87.721774867)
14001409	1168480	1923896	10/18/2025 12:00:00 AM	035XX N CLARK ST	0820	THEFT	6	SIDEWALK	41.94670237	-87.656117076	2025	10/25/2025 03:43:27 PM	41.94670237	-87.656117076	(41.94670237, -87.656117076)
14001262	1176920	1896484	10/18/2025 12:00:00 AM	008XX S WABASH AVE	1320	CRIMINAL DAMAGE	32	PARKING LOT / GAR	41.871295508	-87.625925454	2025	10/25/2025 03:43:27 PM	41.871295508	-87.625925454	(41.871295508, -87.625925454)
14004861	1170792	1921474	10/18/2025 12:00:00 AM	007XX W BELMONT AVE	0820	THEFT	6	APARTMENT	41.940005868	-87.647690125	2025	10/25/2025 03:43:27 PM	41.940005868	-87.647690125	(41.940005868, -87.647690125)
14001368	1199590	1850305	10/18/2025 12:00:00 AM	083XX S BUFFALO AVE	0910	MOTOR VEHICLE THEFT	46	STREET	41.744035687	-87.544254005	2025	10/25/2025 03:43:27 PM	41.744035687	-87.544254005	(41.744035687, -87.544254005)
14001187	1153826	1885018	10/18/2025 12:00:00 AM	028XX S TRUMBULL AVE	0470	PUBLIC PEACE VIOLATION	30	STREET	41.840322451	-87.711017413	2025	10/25/2025 03:43:27 PM	41.840322451	-87.711017413	(41.840322451, -87.711017413)
14001822	1176338	1901346	10/18/2025 12:00:00 AM	001XX N STATE ST	0470	PUBLIC PEACE VIOLATION	32	SIDEWALK	41.884650262	-87.627915459	2025	10/25/2025 03:43:27 PM	41.884650262	-87.627915459	(41.884650262, -87.627915459)
14003180	1169407	1862778	10/18/2025 12:00:00 AM	063XX S RACINE AVE	1340	CRIMINAL DAMAGE	16	CTA TRAIN	41.778969503	-87.654486077	2025	10/25/2025 03:43:27 PM	41.778969503	-87.654486077	(41.778969503, -87.654486077)
14001945	1183166	1871798	10/18/2025 12:00:00 AM	050XX S DREXEL BLVD	0820	THEFT	39	APARTMENT	41.803412024	-87.603764184	2025	10/25/2025 03:43:27 PM	41.803412024	-87.603764184	(41.803412024, -87.603764184)
14003192	1169407	1862778	10/18/2025 12:00:00 AM	063XX S RACINE AVE	1340	CRIMINAL DAMAGE	67	CTA TRAIN	41.778969503	-87.654486077	2025	10/25/2025 03:43:27 PM	41.778969503	-87.654486077	(41.778969503, -87.654486077)

```
SELECT
  MIN(incident_timestamp) AS earliest_date,
  MAX(incident_timestamp) AS latest_date
FROM chicago_final;
```

earliest_date	latest_date
2001-01-01 00:00:00	2025-10-18 00:00:00


```

SELECT
    primary_type,
    COUNT(*) AS total_crimes
FROM chicago_final
GROUP BY primary_type
ORDER BY total_crimes DESC
LIMIT 10;

```

primary_type	total_crimes
THEFT	1789299
BATTERY	1535737
CRIMINAL DAMAGE	958493
NARCOTICS	763916
ASSAULT	564836
OTHER OFFENSE	525604
BURGLARY	445690
MOTOR VEHICLE THEFT	431069
DECEPTIVE PRACTICE	388285
ROBBERY	314679

Step 4: Export Hive Data from HDFS to Local Machine

This step involves extracting the dataset from HDFS to a CSV format, allowing it to be used in Excel and Power BI. We will create smaller, separate files to make it easier and more manageable for Excel and Power BI to work with.

1. In Beeline, use this command to make smaller files: `chicago_locations_2001_2012` and `chicago_locations_2013_2025`.

```

-- 2001 to 2012
INSERT OVERWRITE DIRECTORY
'/user/ksalaz63/chicago_locations_2001_2012'
ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'
WITH SERDEPROPERTIES (
    "separatorChar" = ",",
    "quoteChar" = "\"",
    "escapeChar" = "\\"
)
SELECT
    id, case_number, incident_timestamp, block, primary_type,
    description,
    location_description, arrest, domestic, beat, district, ward,
    community_area, year, latitude, longitude
FROM chicago_final
WHERE year BETWEEN 2001 AND 2012
    AND latitude IS NOT NULL AND longitude IS NOT NULL;

```

```
-- 2013 to 2025
INSERT OVERWRITE DIRECTORY
'/user/ksalaz63/chicago_locations_2013_2025'
ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'
WITH SERDEPROPERTIES (
  "separatorChar" = ",",
  "quoteChar" = "\"",
  "escapeChar" = "\\"
)
SELECT
  id, case_number, incident_timestamp, block, primary_type,
  description,
  location_description, arrest, domestic, beat, district, ward,
  community_area, year, latitude, longitude
FROM chicago_final
WHERE year BETWEEN 2013 AND 2025
AND latitude IS NOT NULL AND longitude IS NOT NULL;
```

2. Use these commands to export `chicago_locations_2001_2012`, use these commands (Git Bash on cluster):

```
$ hdfs dfs -ls /user/ksalaz63/chicago_locations_2001_2012

$ hdfs dfs -get /user/ksalaz63/chicago_locations_2001_2012 ~/chicago_project/

$ ls -lh ~/chicago_project/chicago_locations_2001_2012

To download :

scp
ksalaz63@129.153.113.98:/home/ksalaz63/chicago_project/chicago_locations_2001_2012/00
0000_0 "C:\Users\kesal\Downloads\chicago_locations_2001_2012.csv"
```

3. Use these commands to export `chicago_locations_2013_2025`, use these commands (Git Bash on cluster):

```
$ hdfs dfs -ls /user/ksalaz63/chicago_locations_2013_2025

$ hdfs dfs -get /user/ksalaz63/chicago_locations_2013_2025 ~/chicago_project/

$ ls -lh ~/chicago_project/chicago_locations_2013_2025

To download:

scp
```

```
ksalaz63@129.153.113.98:/home/ksalaz63/chicago_project/chicago_locations_2013_2025/00000_0 "C:\Users\kesal\Downloads\chicago_locations_2013_2025.csv"
```

4. To create `chicago_summary_by_year_type`, use this (Beeline):

```
INSERT OVERWRITE DIRECTORY
'/user/ksalaz63/chicago_summary_by_year_type'
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
SELECT
    year,
    primary_type,
    COUNT(*) AS total_crimes
FROM chicago_final
GROUP BY year, primary_type
ORDER BY year, total_crimes DESC;
```

5. To export `chicago_summary_by_year_type`, use these commands (Git Bash on cluster):

```
$ hdfs dfs -ls /user/ksalaz63/chicago_summary_by_year_type

$ hdfs dfs -get /user/ksalaz63/chicago_summary_by_year_type ~/chicago_project/

$ ls -lh ~/chicago_project/chicago_summary_by_year_type

To download :

scp
ksalaz63@129.153.113.98:/home/ksalaz63/chicago_project/chicago_summary_by_year_type
/000000_0 "C:\Users\kesal\Downloads\chicago_summary_by_year_type.csv"
```

6. To create `chicago_daily_hourly_trends`, use these commands (Beeline):

```
INSERT OVERWRITE DIRECTORY
'/user/ksalaz63/chicago_daily_hourly_trends_data'
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
SELECT
    dayofweek(incident_timestamp) AS day_of_week_num,
    hour(incident_timestamp) AS hour_of_day,
    COUNT(*) AS total_crimes
FROM chicago_final
GROUP BY dayofweek(incident_timestamp), hour(incident_timestamp)
ORDER BY day_of_week_num, hour_of_day;
```

7. To export `chicago_daily_hourly_trends`, use these commands (Git Bash on cluster):

```
$ hdfs dfs -get /user/ksalaz63/chicago_daily_hourly_trends_data ~/chicago_project/

echo "day_of_week_num,hour_of_day,total_crimes" > combined_daily_hourly_trends.csv

cat ~/chicago_project/chicago_daily_hourly_trends_data/000000_0 >>
combined_daily_hourly_trends.csv
```

To download:

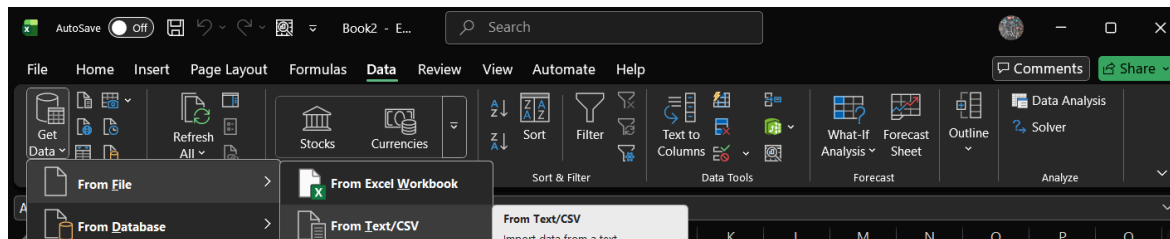
```
scp ksalaz63@129.153.113.98:/home/ksalaz63/combined_daily_hourly_trends.csv
"C:\Users\kesal\Downloads\chicago_daily_hourly_trends.csv"
```

Step 5: Create Visualizations in Excel and Power BI

This step involves creating visualizations from the cleaned files “chicago_locations_2001_2012”, “chicago_locations_2013_2025”, “chicago_summary_by_year_type”, “chicago_trends_district”, and “chicago_locations_clean” using Excel 3D map and Power BI.

Historical Chicago Crime Distribution (2001-2012)- Excel 3D Map

1. To use it on Excel, click on New and on New Workbook, then click Data → Get Data → From File → From Text/CSV, and click ‘chicago_locations_2001_2012.csv’



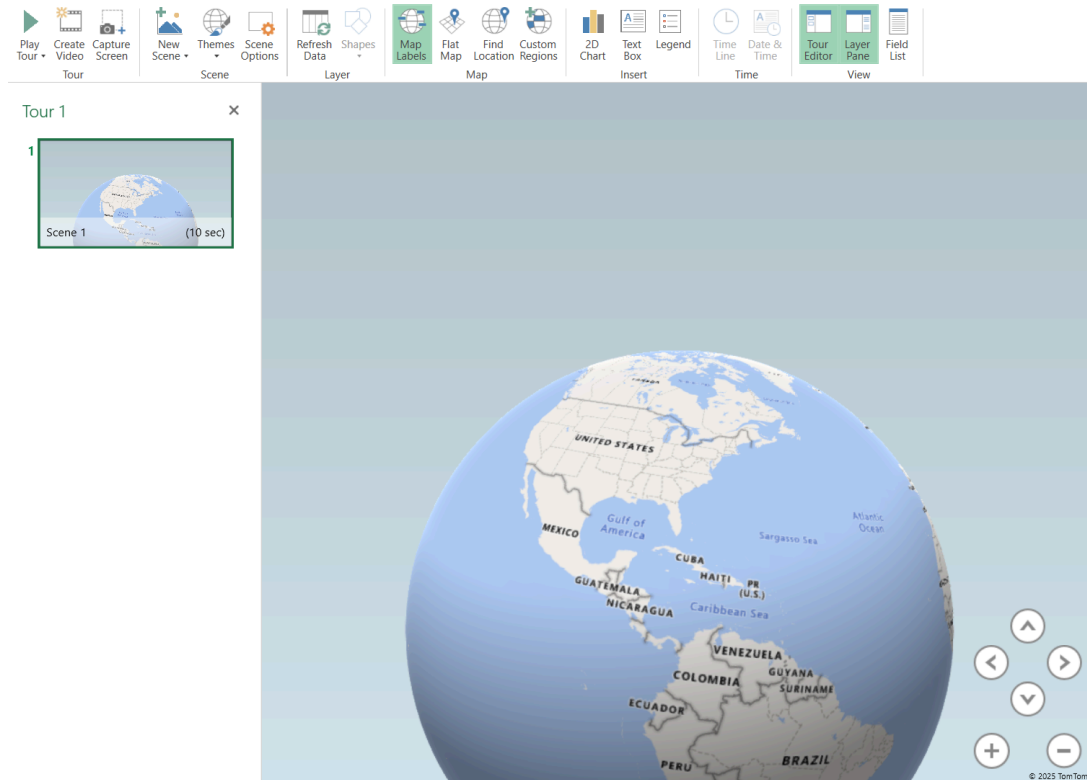
2. On the Delimiter, change it to a comma, and the data will be in the proper columns.
3. Rename the columns in the dataset to be:
ID, Case Number, Incident Timestamp, Block, Primary Type, Description, Location Description, Arrest, Domestic, Beat, District, Ward, Community Area, Year, Latitude, and Longitude.

ID	Case Num	Incident Times	Block	Primary Type	Description	Location Description
8953937	HW102973	12/31/2012 23:58	037XX N NORA AVE	BURGLARY	FORCIBLE ENTRY	RESIDENCE-GARAGE
8950918	HW100021	12/31/2012 23:55	035XX W MONTROSE AVE	BURGLARY	FORCIBLE ENTRY	OTHER
8954299	HW100700	12/31/2012 23:55	058XX S MARYLAND AVE	THEFT	FROM BUILDING	HOSPITAL BUILDING/GROUNDS
8950836	HW100039	12/31/2012 23:55	000XX E OHIO ST	PUBLIC PEACE VIOLATION	OTHER VIOLATION	SIDEWALK
8951459	HW100757	12/31/2012 23:50	028XX N HALSTED ST	THEFT	FROM BUILDING	RESIDENCE
8951048	HV623529	12/31/2012 23:45	070XX S ELIZABETH ST	BATTERY	DOMESTIC BATTERY SIMPLE	APARTMENT
8951016	HW100012	12/31/2012 23:45	026XX N CALIFORNIA AVE	BATTERY	DOMESTIC BATTERY SIMPLE	APARTMENT
8951044	HW100001	12/31/2012 23:43	063XX S PAULINA ST	ASSAULT	AGGRAVATED PO: HANDGUN	STREET
8950817	HV623521	12/31/2012 23:40	043XX S KEDZIE AVE	NARCOTICS	POSS: COCAINE	SIDEWALK
8950828	HW100002	12/31/2012 23:40	022XX N AVERS AVE	BATTERY	AGG: HANDS/FIST/FEET SERIOUS INJURY	APARTMENT
8950880	HW100077	12/31/2012 23:38	062XX S RHODES AVE	CRIMINAL TRESPASS	TO VEHICLE	ALLEY
8950833	HW100013	12/31/2012 23:35	005XX S COLUMBUS DR	BATTERY	SIMPLE	STREET
8950838	HV623527	12/31/2012 23:31	021XX W BELMONT AVE	CRIMINAL DAMAGE	TO PROPERTY	COMMERCIAL / BUSINESS OFFICE
8951353	HW100659	12/31/2012 23:30	001XX W HUBBARD ST	THEFT	POCKET-PICKING	BAR OR TAVERN
8950809	HV623523	12/31/2012 23:30	066XX S STEWART AVE	BATTERY	DOMESTIC BATTERY SIMPLE	APARTMENT
8958717	HW107984	12/31/2012 23:30	052XX S LECLAIRE AVE	THEFT	OVER \$500	STREET
8951370	HW100615	12/31/2012 23:30	056XX S THROOP ST	BATTERY	DOMESTIC BATTERY SIMPLE	APARTMENT
8950819	HV623517	12/31/2012 23:30	056XX W DIVISION ST	BATTERY	DOMESTIC BATTERY SIMPLE	STREET
8951323	HW100597	12/31/2012 23:30	012XX N STATE PKWY	BATTERY	AGG: HANDS/FIST/FEET SERIOUS INJURY	SIDEWALK
8950832	HW100023	12/31/2012 23:30	020XX N HALSTED ST	THEFT	FROM BUILDING	RESTAURANT
8950977	HV623519	12/31/2012 23:25	011XX S CENTRAL PARK AVE	CRIMINAL DAMAGE	TO VEHICLE	STREET
9169120	HV623504	12/31/2012 23:16	085XX S KILPATRICK AVE	OFFENSE INVOLVING CHILDREN	AGG CRIM SEX ABUSE FAM MEMBER	RESIDENCE
8950881	HW100136	12/31/2012 23:15	114XX S DR MARTIN LUTHER KING JR DR	NARCOTICS	POSS: HALLUCINOGENS	RESIDENCE PORCH/HALLWAY
8951354	HW100638	12/31/2012 23:15	076XX S MICHIGAN AVE	MOTOR VEHICLE THEFT	AUTOMOBILE	STREET
8951442	HW100695	12/31/2012 23:00	033XX W FULTON BLVD	CRIMINAL DAMAGE	TO PROPERTY	APARTMENT
8951709	HW101116	12/31/2012 23:00	009XX N LEAMINGTON AVE	THEFT	\$500 AND UNDER	RESIDENCE
8963093	HW100055	12/31/2012 23:00	026XX W BLOOMINGDALE AVE	CRIMINAL DAMAGE	TO VEHICLE	STREET
8950864	HV623531	12/31/2012 23:00	022XX S MICHIGAN AVE	OTHER OFFENSE	LICENSE VIOLATION	ABANDONED BUILDING
8950839	HV623522	12/31/2012 23:00	023XX W MCLEAN AVE	THEFT	\$500 AND UNDER	RESIDENCE
8953020	HW102159	12/31/2012 23:00	102XX S EWING AVE	DECEPTIVE PRACTICE	CREDIT CARD FRAUD	BAR OR TAVERN
8950803	HV623514	12/31/2012 23:00	000XX E ROOSEVELT RD	DECEPTIVE PRACTICE	THEFT OF LABOR/SERVICES	OTA PLATFORM
8950821	HW100037	12/31/2012 23:00	007XX N ELIZABETH ST	THEFT	\$500 AND UNDER	VEHICLE NON-COMMERCIAL
8950812	HV632518	12/31/2012 22:58	072XX S CHAMPLAIN AVE	CRIMINAL DAMAGE	TO PROPERTY	APARTMENT
8950855	HV623500	12/31/2012 22:53	105XX S CORLISS AVE	BATTERY	DOMESTIC BATTERY SIMPLE	RESIDENCE

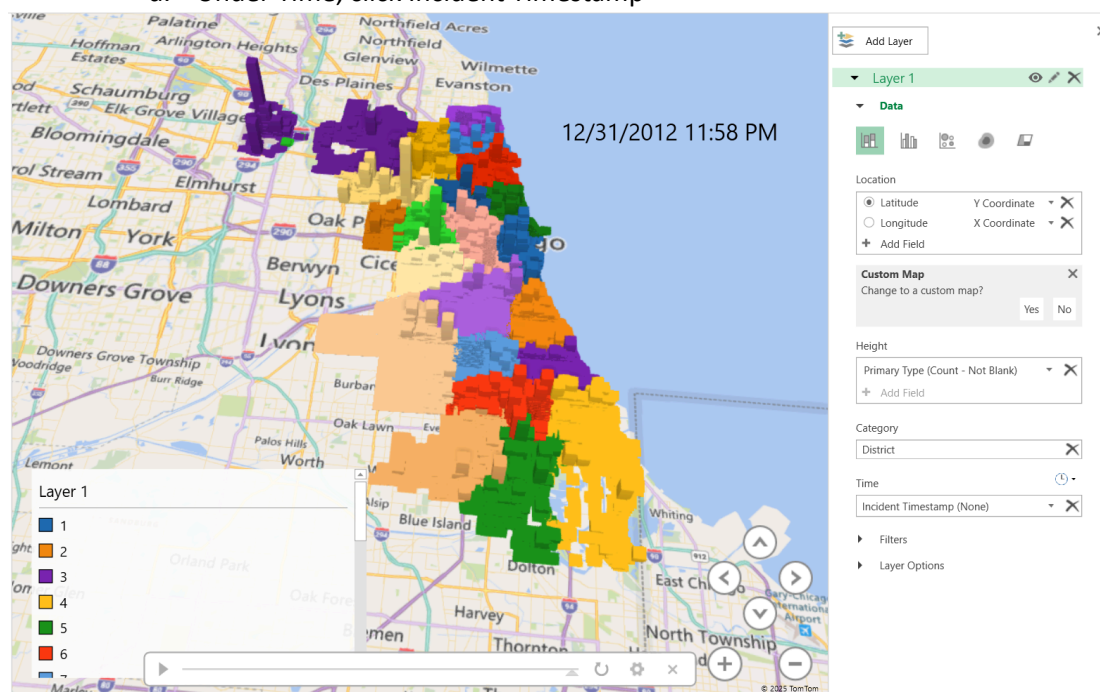
4. Select all data (Ctrl + A) and search up 3D map

The screenshot shows the Microsoft Excel interface. The 'Home' tab is active. A search bar is open, displaying 'chicago_lo...' and a filter of '8953937'. The '3D Map' panel is open on the right, showing a search bar and a list of 'Recently Used Actions' including '3D Map', 'Open 3D Maps', and 'Analyze Data'. The 'Suggested Actions' panel is also visible, listing 'Freeze Panes', 'Header and Footer', and 'Insert Sheet Rows'.

5. The 3D map will be empty, and we will want to click the map labels:

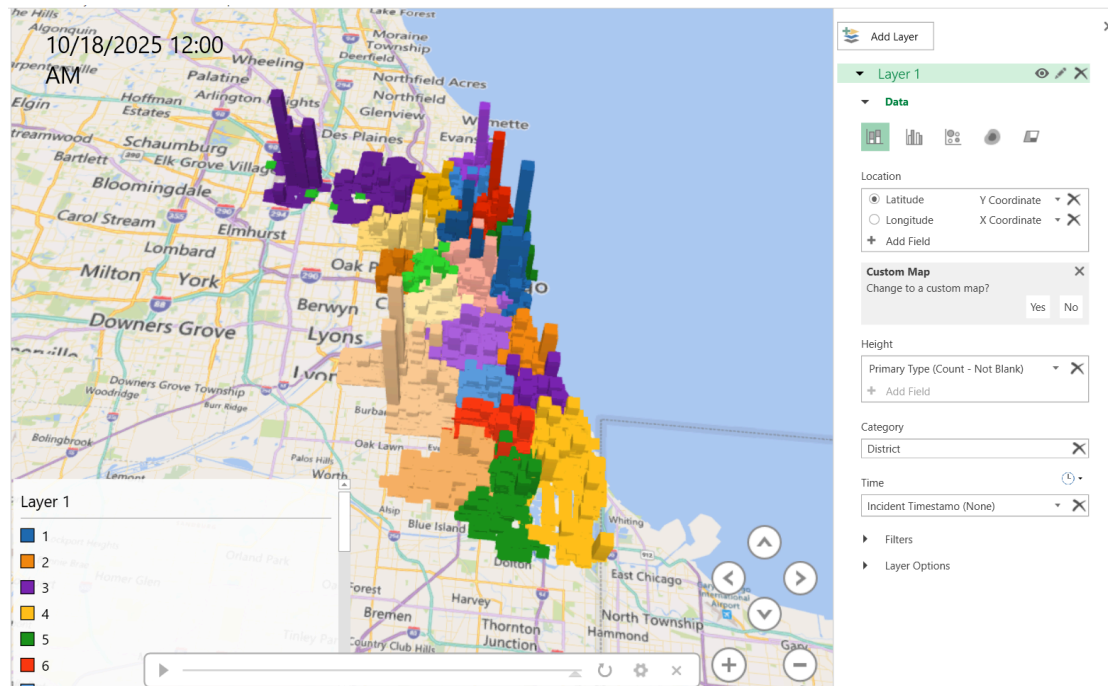


6. For the first layer, click the stacked column
 - a. Under Location, add Latitude (Y Coordinate) and Longitude (X Coordinate)
 - b. Under Height, add Primary Type (Count - Not Blank)
 - c. Under Category, click District.
 - d. Under Time, click Incident Timestamp



Historical Chicago Crime Distribution (2013-2025)- Excel 3D Map

- Repeat steps 1-6 from the previous visual to create the crime distribution for the years 2013-2025, using the file 'chicago_locations_2013_2025.csv'.
 - You can even use the 'Play Tour' button on the top left to see the progression



Line Chart- Trends of Top 10 Chicago Crime Types

- Using Power BI, click + New Report and then click Get Data. Using the upload file feature, upload the file 'chicago_summary_by_year_type'. Wait for it to download and click 'Next'.
- In the Preview file data, make sure the delimiter is 'Comma' and click 'Next', with your data looking like this:

Preview file data

URL: https://csula-my.sharepoint.com/personal/ksalaz63_calstatela_edu/Documents/Apps/Microsoft Power Query/Uploaded Files/chicago_summary_by_year_type 1.csv

File origin	Delimiter	Data type detection
65001: Unicode (UTF-8)	Comma	Based on first 200 rows
Column1	Column2	Column3
2001	THEFT	99291
2001	BATTERY	93471
2001	CRIMINAL DAMAGE	55865
2001	NARCOTICS	50568
2001	ASSAULT	31388
2001	OTHER OFFENSE	29669
2001	MOTOR VEHICLE THEFT	27556
2001	BURGLARY	26014
2001	ROBBERY	18441
2001	DECEPTIVE PRACTICE	14958
2001	CRIMINAL TRESPASS	13242
2001	PROSTITUTION	6026
2001	WEAPONS VIOLATION	4278
2001	PUBLIC PEACE VIOLATION	2750
2001	OFFENSE INVOLVING CHILDREN	2268

- Change the Column names to Year, Primary Type, and Total Crimes. Click 'Create a report' after renaming the columns and name it 'chicago_summary_by_year_type':

Queries [1] < Table.RenameColumns("#Changed column type", [{"Column1", "Year"}, {"Column2", "Primary type"}, {"Column3", "Total Crimes"}])

123 Year	123 Primary type	123 Total Crimes
1	2001 THEFT	99291
2	2001 BATTERY	93471
3	2001 CRIMINAL DAMAGE	55865
4	2001 NARCOTICS	50568
5	2001 ASSAULT	31388
6	2001 OTHER OFFENSE	29669
7	2001 MOTOR VEHICLE THEFT	27356
8	2001 BURGLARY	26014
9	2001 ROBBERY	18441
10	2001 DECEPTIVE PRACTICE	14958
11	2001 CRIMINAL TRESPASS	13242
12	2001 PROSTITUTION	6026
13	2001 WEAPONS VIOLATION	4278
14	2001 PUBLIC PEACE VIOLATION	2750
15	2001 OFFENSE INVOLVING CHILDREN	2268
16	2001 SEX OFFENSE	2238
17	2001 CRIM SEXUAL ASSAULT	1771
18	2001 LIQUOR LAW VIOLATION	1637
19	2001 ARSON	1011
20	2001 GAMBLING	934
21	2001 KIDNAPPING	933
22	2001 HOMICIDE	667
23	2001 INTERFERENCE WITH PUBLIC OFFI...	406
24	2001 INTIMIDATION	280
25	2001 STALKING	204

Columns: 3 Rows: 99+

Query settings >

Properties

Name

chicago_summary_by_ye...

Applied steps

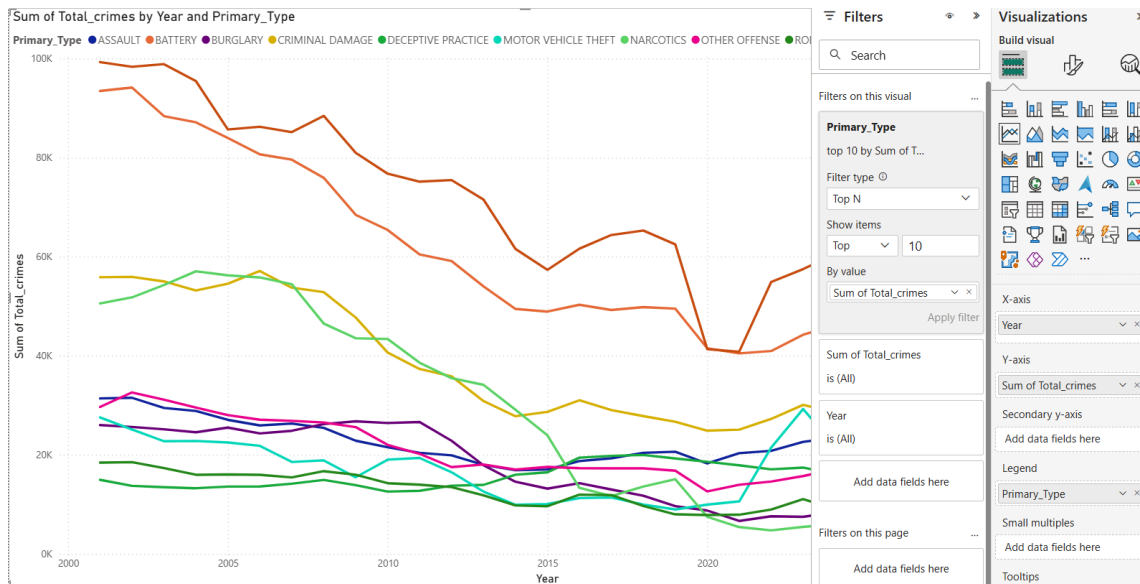
Source

Changed c...

Renamed c...

Create a report Cancel

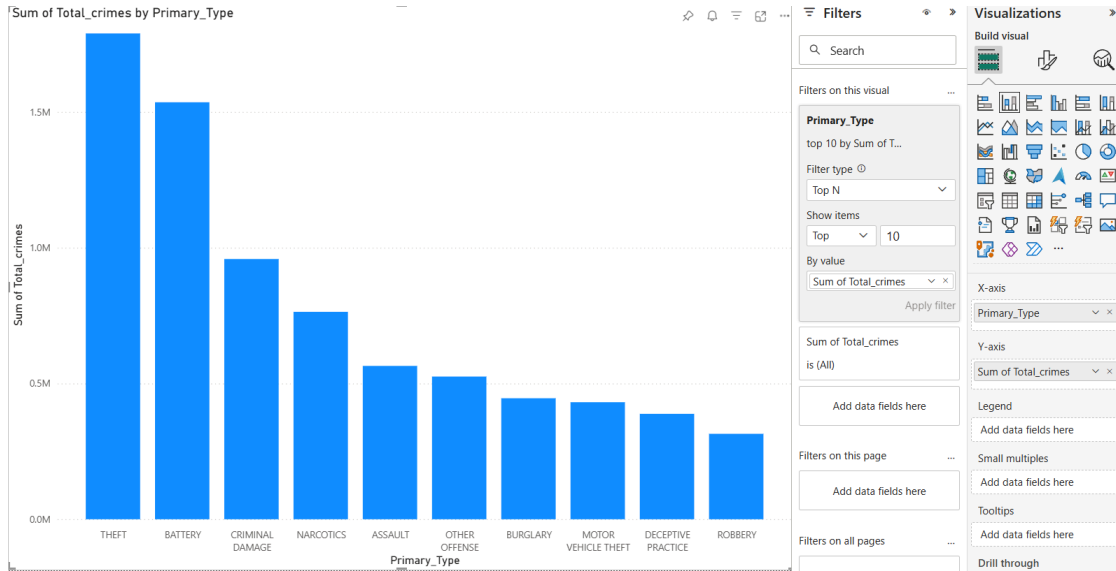
- You will see an empty page. In the Visualization section, click 'Line Chart'. In the data section, drag 'Year' to the X-axis and 'Total crimes' to the Y-axis (Sum of Total_crimes).
 - In the filters section, click the down arrow (next to the Lock symbol) on the Primary_Type, for the Filter type select Top N, and in the show items section, write 10. In the By Value section, drag the Total_crimes to that section and click Apply Filter. It will make a visual like this



Stacked Column Chart- Sum of Count By Crime Type

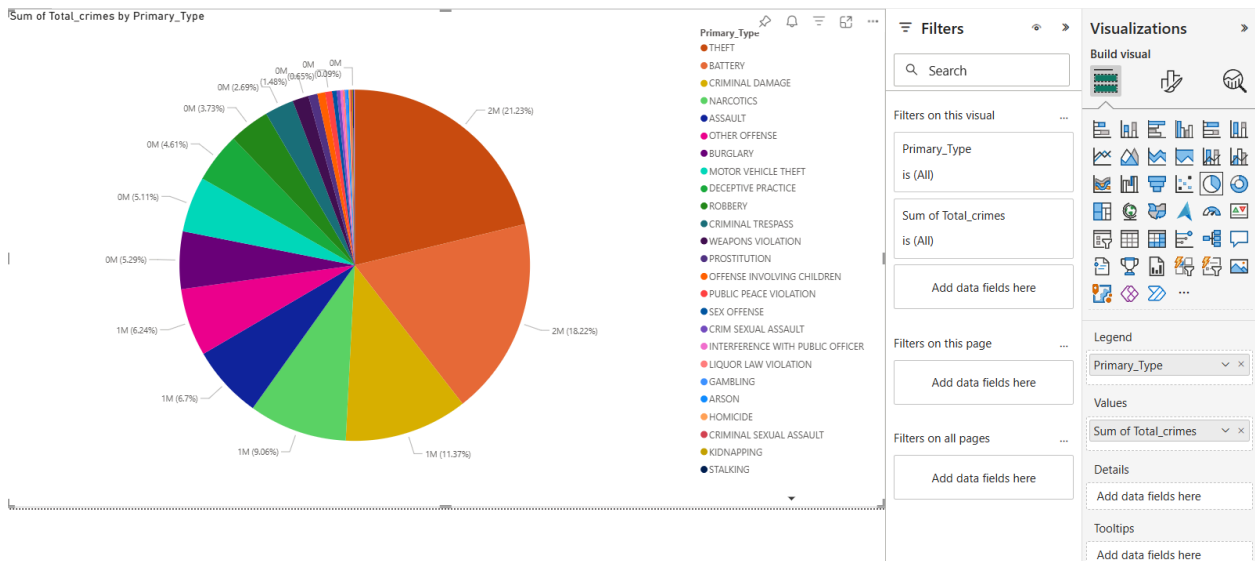
- Make a new page. In the Visualization section, click 'Stacked Column Chart'. In the data section, drag 'Primary Type' to the X-axis and 'Total crimes' to the Y-axis (Sum of Total_crimes).

- a. In the filters section, click the down arrow (next to the Lock symbol) on the Primary_Type, for the Filter type select Top N, and in the show items section, write 10. In the By Value section, drag the Total_crimes to that section and click Apply Filter. It will make a visual like this:



Pie Chart- Crime Distribution by Type

6. Make a new page. In the Visualization section, click 'Pie Chart'. In the data section, drag 'Primary Type' to the Legend and 'Total crimes' to the Values (Sum of Total_crimes).
 - a. It will create a visual that looks like this:



Line Chart- 24-Hour Crime Pattern

1. Repeat steps 1-2 of the previous visual to create a Stacked Column chart visual with the file 'chicago_daily_hourly_trends.csv'

2. In the Power Query view, the Column will already be named and will look like this:
 - a. Click Next and click Create Report, where you will name it 'chicago_daily_hourly_trends'

Power Query

Get data

Preview file data

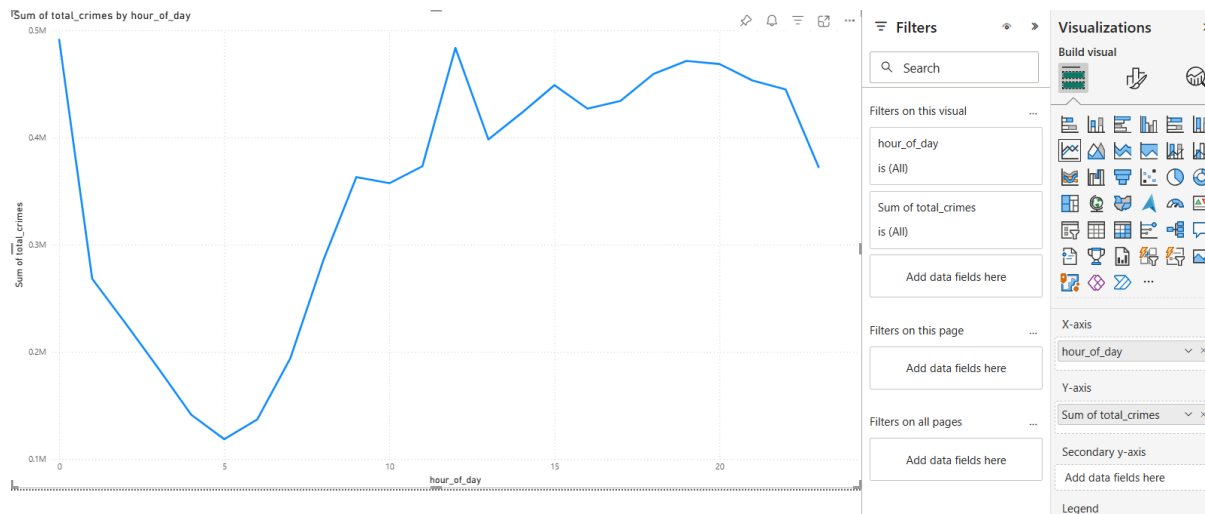
URL: https://csula-my.sharepoint.com/personal/ksalaz63_calstatela_edu/Documents/Apps/Microsoft Power Query/Uploaded Files/chicago_daily_hourly_trends 2.csv

File origin: 65001: Unicode (UTF-8) Delimiter: Comma Data type detection: Based on first 200 rows

day_of_week_num	hour_of_day	total_crimes
1	0	78360
1	1	54466
1	2	49890
1	3	43995
1	4	32850
1	5	23356
1	6	18598
1	7	19280
1	8	26749
1	9	38424
1	10	42261
1	11	45387
1	12	57377
1	13	47905
1	14	50379
1	15	53548
1	16	53685
1	17	55702
1	18	59151
1	19	61959
1	20	63147
1	21	61887
1	22	61231

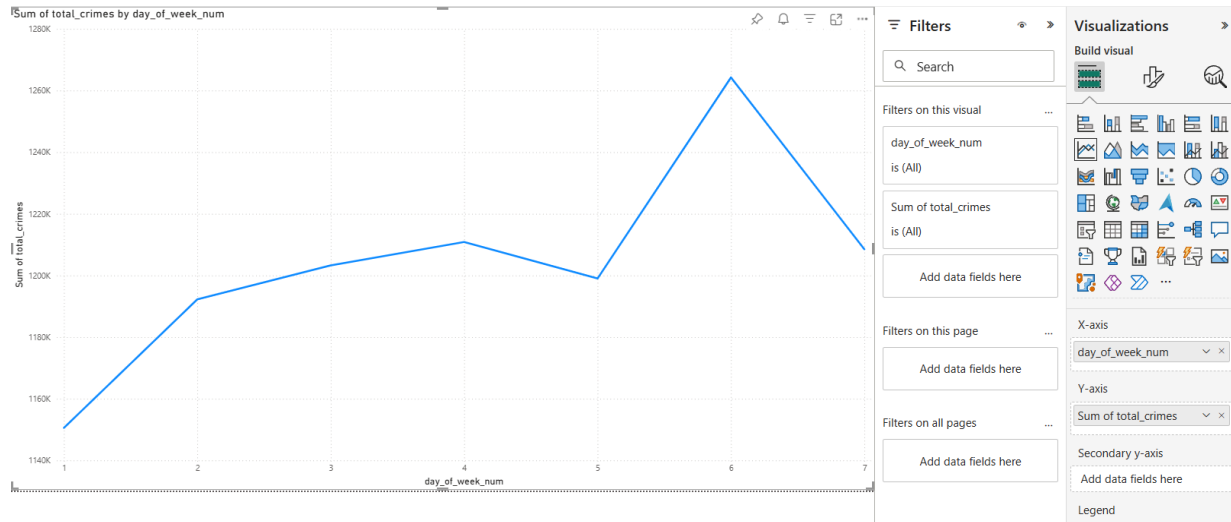
Back Cancel Next

3. You will see an empty page. In the Visualization section, click 'Line Chart'. In the data section, drag 'hour_of_day' to the X-axis and 'Total crimes' to the Y-axis. It will make a visual like this:



Line Chart- Total Crime Count by Day of Week

4. Make a new page by clicking page 2. Click 'Line Chart'. In the data section, drag 'day_of_week_num' to the X-axis and 'Total crimes' to the Y-axis. It will make a visual like this:



References

1. URL of Data Source:

https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-Present/ijzp-q8t2/about_data

2. URL of your GitHub:

<https://github.com/ksalaz63/ChicagoCrime2001-2025>

3. URL of References: